Scheme of Valuation/Answer Key

(Scheme of evaluation (marks in brackets) and answers of problems/key)

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R, S) / S6 (PT) (R, S) Examination May 2024 (2019 Scheme)

Course Code: CST302

Course Name: COMPILER DESIGN

Max. Marks: 100 | Duration: 3 Hours

PART A

		Answer all questions, each carries 3 marks.	Marks
1		Token : Token is a sequence of characters that can be treated as a single logical entity. Pattern : A set of strings in the input for which the same token is produced as output. This set of strings is described by a rule called a pattern associated with the token. Lexeme : A lexeme is a sequence of characters in the source program that is matched by the pattern for a token - 1.5 marks Identifying values from given statement 1.5 marks	(3)
2		Bootstrapping is used to produce a self-hosting compiler. Self-hosting compiler is a type of compiler that can compile its own source code. Bootstrap compiler is used to compile the compiler and then you can use this compiled compiler to compile everything else as well as future versions of itself - 3 marks	(3)
3		Procedure 1.5 Solution 1.5	(3)
4		Procedure 1.5 Solution 1.5	(3)
5		Definition- 2 marks Example – 1 mark	(3)
6		Shift, reduce, accept and error – with description- 3 marks	(3)
7		Structure of activation record.-3 marks	(3)
8		Any three points 3 marks .	(3)
9		DAG and syntax tree for the statement d+a*(b-c) + (b-c)*d- 1.5 marks each	(3)
10		Example code- 1.5 induction variable elimination technique – 1.5	(3)

PART B

Answer one full question from each module, each carries 14 marks.

		Module I	
11	a)	6 phases explanation 5 marks	(8)

		Output of each state for the given code segment 3 markks	
	b)	Transition diagrams of (i)relational operators (ii)identifiers 3 marks each	(6)
		OR	
12	a)	Input buffering with necessary diagrams- 5 marks the advantages of using double buffers and sentinels in input buffering. 3 marks	(8)
	b)	Four compiler construction tools 1.5 marks each.	(6)
		Module II	
13	a)	Left recursion elimination- 2 marks parsing table – 5 marks .Yes ll(1)- 1 mark. LL(1) parsing table is as shown below.	(8)
	b)	The recursive descent parsing procedure – 6 marks	(6)
		OR	
14	a)	Algorithm of FIRST and FOLLOW 6 marks role– they help the parser determine which production rule to apply to the given input. First tells which terminal can start production whereas the follows tells the parser what terminal can follow a non-terminal. 2 marks	(8)
	b)	Three address code of using each method- 2 marks each	(6)
		Module III	
15	a)	Since there is a mistake in the Question (Missing One production) , Students who have attempted the question with the given productions may be given Full Marks after verifying the steps followed for SLR parsing table construction. Otherwise marks can be awarded accordingly.	(8)
	b)	Handle and handle pruning with example 3 marks Listing out the handles 3 marks	(6)
		OR	
16	a)	Constructing sets of items: 3 marks CLR Parsing table: 3 marks	(8)

LL(1) parsing table (Q13 a):

	a	(	)	,	$
S	S->a	S->(L)			
L	L->SL’	L->SL’			
L’			L’-> ε	L’->,SL’	

		Yes. Grammar is CLR with justification : 2 marks	
	b)	Shift reduce conflict and reduce reduce conflict. 3 marks each	(6)
		Module IV	
17	a)	SDD 4 marks, Annotated parse tree and evaluation 4 marks	(8)
	b)	SDD 4 marks, Annotated parse tree and evaluation 4 marks	(6)
		OR	
18	a)	Stack allocation- 3 Heap allocation 3 Static allocation 2	(8)
	b)	3 representation method and solution 2 marks each	(6)
		Module V	
19	a)	Any four techniques in function preserving transformations and loop optimizations- each carries 2 marks	(8)
	b)	Basic block - 2 marks Four Structure preserving transformations on a basic block- 4 marks	(6)
		OR	
20	a)	Any four Peephole optimization techniques with example – 2 marks each	(8)
	b)	Any three issues in the design of a code generator 1. Input to code generator 2. target program 3. memory management 4. instruction selection 5. register allocation 6. evaluation order	(6)
		****	